Semi-focus

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Earlier this month, there was a flurry of news around Apple's list of concessions requested from the Indian government. This was about them setting up assembly units in the country. The list of concessions turned out to be a little over the top which included a reported 15-year waiver of customs duty on iPhone kit imports and a few other favourable policies for sourcing locally. Obviously, no company would set up shop in a country which wasn't aligned with their business goals, let along a company like Apple which is known for its exorbitantly high mark-up. Moreover, India does relax norms provided companies source locally but these are limited to the initial three years and if the FDI is above 51 per cent, then further relaxation is applied for another five years. The waiver sought by Apple is for 15 years. And that too for completely knocked down (CKD) and semi-knocked down (SKD) units. That is, Apple would be importing manufactured components and just assembling them here. There's no actual making in India happening.

There is a strong correlation between manufacturing and a country's economy. You can add the obligatory 'duh!' here. And we don't need to look far to know how big an impact this has. China's markets are favoured towards indigenous semiconductor design which is why you have a lot of firms investing heavily in Chinese semiconductor fabs. While the Indian government is doing the right thing with the whole Make In India initiative, they aren't targeting the right kind of companies. Mobile phone companies are all fine and dandy but they still look towards China to get their SoCs, NAND chips and practically everything under the sun that has electrons flowing through them. There's one fab in the country owned by ISRO which only has 180 nm transistor technology. That's as good as not having a fab. In case you haven't been keeping up with the industry news, most smartphones have SoCs which are using 28 nm transistors and fast moving budget smartphones have already started using new SoCs with 14 nm transistors.

There's hope on the horizon though. Later this year, we're going to have our very first fab in Gujarat. And the best part? Hindustan Semiconductor Manufacturing Corporation (HSMC) will start with 90, 65 and 45 nm process nodes and then move on to smaller nodes i.e 28 and 22 nm in its second phase. No doubt, it will be some time before mainstream companies opt to have their silicon manufactured at HSMC as large transistors wreak havoc on battery life. And with the IoT industry already blowing up, it is absolutely essential that small transistors be focused on, as that's what the entire industry is using. Starting off with 90 nm transistors which were popular back in 2004 is a wrong move. You can pack fewer transistors in a given area, they run hotter and they consume more power. And it's not like HSMC is completely in the dark, the fab is being built with the help of STMicroelectronics and Silterra, two well established entities in the industry.

There are plenty of fabs all over the world which are manufacturing 90 nm wafers, and jumping in the midst of that isn't going to bode well. Especially, if you wish to earn from 90 nm and then progress towards smaller process nodes based on natural growth. That's going to be a slow and painful climb. So what works? Semiconductor designing. As an industry poised to compete intensely in a race to be smaller, more powerful and more efficient, there's a lot happening with VLSI and chip design. There are very few companies doing actual R&D while the rest of the world piggy backs on their design. This is the foundation of the semiconductor industry and that's where our focus should be. Build original IPs and the fabs will follow and so will the companies.

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